added to the atmosphere by human activity. However, the gas alters the oceans' chemistry.

In 2003, American climate scientists Ken Caldeira and Michael E. Wickett investigated the effects of CO_2 pollution on the oceans. They took samples of seawater from around the world, and found that the acidity had increased measurably in the past 200 years of industrialization. They coined the term "ocean acidification" and predicted that this change could accelerate over the next 50 years, with damaging results.

Many sea creatures rely on the natural alkalinity of seawater to maintain carbonates for building their shells and skeletons. Even a slight decrease in alkalinity seriously disrupts growth, especially for sensitive creatures such as corals and plankton. Acidification might wipe out corals within decades; if they go, so do the reef ecosystems. Phytoplankton are the foundation of the ocean food web, and are vital to maintaining global oxygen levels.

Ocean acidification is far harder to reverse than the atmospheric effects of CO_2 emissions, and its devastating impact on biodiversity, fisheries, and food security remains a serious concern.

"Most carbon dioxide released into the atmosphere as a result of burning fossil fuels will be absorbed by the ocean."

Ken Caldeira and Michael Wickett

See also: Global warming • Pollution • Endangered habitats • Acid rain • Halting climate change

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